



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

First Semester 2026-27
COURSE HANDOUT PART-II

Date: 01.08.2026

In addition to part I (General handout for all courses appended to the timetable) this portion gives further details regarding the course.

Course No. : CHEM F330
Course Title : Photophysical Chemistry
Instructor-in-charge : INAMUR RAHAMAN LASKAR

1. Course Description:

Absorption of the electromagnetic radiation; photophysical processes such as fluorescence, phosphorescence, non-radiative transitions, and delayed luminescence, excimer and exciplex formation; triplet state: radiative and non-radiative transitions; energy transfer, fluorescence resonance energy transfer (FRET), quenching of fluorescence; fluorescence decay; protein and DNA fluorescence; time-resolved emission spectra (TRES); time-dependent anisotropy decays; application of photophysics for the characterization of biological and bio-mimicking systems. In addition to the theory, through simple experiments, laboratory training will be imparted.

2. Scope and Objective of the Course: The main objective of this course is to introduce the students to basic concept of physical processes induced by absorption of light, pathways for loss of electronic excitation such as fluorescence, quenching, and energy transfer.

3. Course learning outcomes

Key concepts of photophysical processes, analysis of Jablonski diagrams and excited state dynamics

Interpretation and applications of spectroscopic techniques

Calculation of photophysical parameters

Evaluation of mechanisms of energy and electron transfer

Application of photophysical principles to real-world applications

Design and critique experiments in photophysical chemistry

4. Text Books and Reference books:

TB. Principles of Fluorescence Spectroscopy by Joseph R. Lakowicz, Third Edition, Springer

RB1. Modern Molecular Photochemistry by N. J. Turro, Benjamin/Cummings Pub. Co.

5. Method of conduct of the course: 14 weeks with 3 lectures per week and five lab classes.

6. Course Plan:

Week No.	Module	Lect No.	Reference	Learning outcomes
1-3	Introduction to Fluorescence: Phenomena of Fluorescence, Jablonski Diagram; Characteristics of Fluorescence Emission, Stokes Shift, Exceptions to the Mirror-Image Rule; Fluorescence Lifetimes and Quantum Yields; Fluorescence Quenching, Timescale of Molecular Processes in Solution; Fluorescence Anisotropy, Resonance Energy Transfer, Steady-	1-8	TB: 1.1-1.10	Define fluorescence and distinguish it from other photophysical processes such as phosphorescence and delayed luminescence



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

	State and Time-Resolved Fluorescence; Biochemical Fluorophores Fluorescent Indicators Molecular information from Fluorescence, Biochemical Examples of Basic Phenomena			
4-5	Instrumentation for Fluorescence Spectroscopy: Spectrofluorometers, Light Sources, Monochromators; Optical Filters and Signal Purity, Photomultiplier Tubes; Polarizers, Corrected Excitation Spectra, Corrected Emission Spectra; Quantum Yield Standards, Effects of Sample Geometry, Common Errors in Sample Preparation, Absorption of Light and Deviation from the Beer-Lambert Law	9-14	TB: 2.1-2.13	Describe the key components of a fluorescence spectrometer, light sources, monochromators, detectors, and sample holders and how these properties affect the observed spectral data
5-6	Solvent and Environmental Effects on fluorescence: Overview of Solvent Polarity Effects, Effects of Solvent Polarity, Polarity Surrounding a Membrane-Bound Fluorophore, other Mechanisms for Spectral Shifts; General Solvent Effects: The Lippert-Mataga Equation, Derivation of the Lippert Equation, Application of the Lippert Equation; Specific Solvent Effects, Specific Solvent Effects and Lippert Plots, Temperature Effects, Phase Transitions in Membranes; Additional Factors that Affect Emission Spectra, Locally Excited and Internal Charge-Transfer States, Excited- State Intramolecular Proton Transfer (ESIPT), Changes in the Non-Radiative Decay Rates, Changes in the Rate of Radiative Decay, Effects of Viscosity, Probe-Probe Interactions	15-20	TB: 6.1-6.8	Explain how solvent polarity influences fluorescence spectra, describe the effects of solvent viscosity and temperature on fluorescence intensity and lifetime. Understand the role of hydrogen bonding and specific solvent-solute interactions
7	Dynamics of Solvent and Spectral Relaxation: Overview of Excited-State Processes, Time-Resolved Emission Spectra; Measurement of Time-Resolved Emission Spectra	21-23	TB: 7.1-7.4	Describe the concept of spectral relaxation and its relation to solvent dynamics around an excited fluorophore.



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

	(TRES), Direct Recording of TRES, TRES from Wavelength-Dependent Decays			Explain how solvent molecules reorganize around excited states and the timescales involved in this relaxation process.
8-9	Quenching of Fluorescence: Quenchers of Fluorescence, Theory of Collisional Quenching; Derivation of the Stern-Volmer Equation; Interpretation of the Bimolecular Quenching Constant; Theory of Static Quenching; Combined Dynamic and Static Quenching; Examples of Static and Dynamic Quenching, Deviations from the Stern-Volmer Equation	24-26	TB: 8.1-8.8	Explain the physical and chemical processes responsible for quenching fluorescence. Apply the Stern-Volmer equation to analyze quenching data and determine quenching constants. Identify common quenchers and their modes of interaction with fluorophores.
10	Mechanisms and Dynamics of Fluorescence Quenching: Comparison of Quenching and Resonance Energy Transfer, Distance Dependence of RET and Quenching, Encounter Complexes and Quenching Efficiency; Mechanisms of Quenching, Intersystem Crossing, Electron-Exchange Quenching, Photoinduced Electron Transfer; Energetics of Photoinduced Electron Transfer, Examples of PET Quenching, PET in Linked Donor-Acceptor Pairs PET; Quenching in Biomolecules, Quenching of Indole by Imidazolium, Quenching by DNA Bases and Nucleotides, Single-Molecule PET	27-30	TB: 9.1-9.5	Explain the molecular interactions and physical processes for quenching mechanism. Analyze the kinetics and dynamics of quenching processes, evaluate how quenching affects fluorescence lifetimes and intensities, and interpret time-resolved fluorescence data accordingly. Identify factors influencing quenching efficiency
11	Fluorescence Anisotropy: Definition of Fluorescence Anisotropy, Theory for Anisotropy, Measurement of Fluorescence Anisotropies L-Format and T-Format Magic-Angle Polarizer Conditions, Effect of Resonance Energy Transfer on the Anisotropy, Trivial Causes of Depolarization, Factors Affecting the Anisotropy	31-34	TB: 10.1-10.5	Define fluorescence anisotropy and understand its physical basis related to molecular rotation. Explain how polarized excitation and emission are used to measure anisotropy in fluorophores.



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

13-14	Energy Transfer: Characteristics of Resonance Energy Transfer, Theory of Energy Transfer for a Donor–Acceptor Pair; Orientation Factor, Dependence of the Transfer Rate on Distance (r), the Overlap Integral (J)	35-38	TB: 13.1-13.3	Explain the physical principles and mechanisms underlying non-radiative energy transfer between donor and acceptor molecules. Understand the distance dependence of energy transfer efficiency.
Experiments 1. Fluorescence quantum yield of a fluorophore. 2. Fluorescence quenching of a fluorophore. 3. Fluorescence aggregation-caused quenching. 4. Fluorescence aggregation induced emission. 5. Identification of charge transfer state.				

7. Evaluation Scheme:

Component	Weightage (%)	Duration	Date & Time	Nature of component (Close Book/ Open Book)
Continuous Evaluations	25	15 min. (Each Test)	Continuous	Closed Book
Lab Experiments	5	100 min	Continuous	-
Mid-Semester Test	30	90 Min.	07/10 FN1	TBA
Comprehensive Examination	40	180 Min.	08/12 FN	TBA

7. Minimum criterion for awarding valid grades: Student should obtain 30% of the average of the top five performers in the class, or 40% of the median marks of the class, whichever is lower to clear the course. If any student gets the marks lower than the prescribed standard mentioned above, he/she may be awarded NC.

8. Attendance Policy: A total five tutorial tests will be conducted; One of the tests will be linked to 50 % class attendance

9. Chamber Consultation Hour: Friday 4.00-5.00 pm

10. Notices: Notices, if any, will be displayed on **Nalanda (or notified by email)**.

11. Make-up Policy: Make-up would be considered only for **genuine cases**.

INAMUR RAHAMAN LASKAR
 Instructor-in-charge



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

Course No. CHEM F330